Piecewise Self-Similar Radiation-Hydrodynamics Analytic Profiles

Antigravity & Tal Ramot

May 2026

1 Introduction

This document summarizes the complete piecewise self-similar analytical profiles for the radiation-hydrodynamics simulation under a constant boundary temperature drive of $T(0,t) = 1\,\mathrm{HeV}$ ($\tau = 0$) in a gold-like medium ($\rho_0 = 19.32\,\mathrm{g/cm^3}$). The system contains two distinct physical domains separated by the ablation front $m_f(t)$:

1. **Subsonic Ablation Regime ($m < m_f(t)$):** A radiation-diffusion-driven heat wave with low density and high temperature.
2. **Shock/Piston Regime ($m > m_f(t)$):** A hydrodynamically driven shock wave where radiation is negligible, resulting in high compressed density and high pressure.

The solutions are patched at $m = m_f(t)$ by matching the pressure and velocity. We present the explicit formulas for each profile (p, u, ρ, T) as functions of the Lagrangian mass coordinate m and time t.

2 Wave Front and Mass Coordinates

All physical quantities are scaled using nanoseconds (ns) for time, $\mathrm{g/cm^2}$ for mass coordinates, $\mathrm{g/cm^3}$ for density, MBar for pressure, km/s for velocity, and HeV for temperature.

2.1 Ablation (Heat) Front

The ablation front in similarity coordinates is $\xi_f \approx 0.427735$. In physical coordinates, the total ablated mass grows as a power law in time:

$$m_f(t) = m_{f0}\left(\frac{t}{\mathrm{ns}}\right)^{0.515625} \approx 0.00767554\left(\frac{t}{\mathrm{ns}}\right)^{0.515625}\ \mathrm{g/cm^2} \tag{1}$$

2.2 Shock Front

The shock front mass in similarity coordinates is $\xi_s \approx 1.669247$. The total shocked mass scales as:

$$m_s(t) = m_{s0}\left(\frac{t}{\mathrm{ns}}\right)^{0.7765} \approx 0.0120786\left(\frac{t}{\mathrm{ns}}\right)^{0.7765}\ \mathrm{g/cm^2} \tag{2}$$

3 Explicit Piecewise Profiles

3.1 Subsonic Ablation Region ($m < m_f(t)$)

In this region, the similarity variable is normalized to:

$$y = \frac{m}{m_f(t)} \in [0, 1] \tag{3}$$

Density Profile $\rho(m,t)$: The bulk density profile is fitted using a singular power law near the front:

$$\rho(m,t) \approx 0.173930 \left(\frac{t}{\text{ns}}\right)^{-0.520833} (1-y)^{-0.491367}\,\text{g/cm}^3 \tag{4}$$

Pressure Profile $p(m,t)$: The pressure scales as:

$$p(m,t) \approx 7.051327 \left(\frac{t}{\text{ns}}\right)^{-0.447917} \left[0.348591 y^{0.876766} + 0.029050 y^{20.948361}\right]\,\text{MBar} \tag{5}$$

Velocity Profile $u(m,t)$: We provide two alternative high-accuracy fits for the velocity profile:

- **Rational Fit ($R^2 = 0.9689$):**

$$u(m,t) \approx -202.645 \left(\frac{t}{\text{ns}}\right)^{0.036458} \frac{1-y}{1+5.48441y}\,\text{km/s} \tag{6}$$

- **Rational Fractional Fit ($R^2 = 0.9849$):**

$$u(m,t) \approx -289.838 \left(\frac{t}{\text{ns}}\right)^{0.036458} \frac{1-y^{0.2340}}{1+0.7524y}\,\text{km/s} \tag{7}$$

Temperature Profile $T(m,t)$: The temperature profile satisfies:

$$T(m,t) \approx 1.011087(1-y)^{0.241582}\,\text{HeV} \tag{8}$$

Note that at the boundary $m = 0$, $T(0,t) \approx 1\,\text{HeV}$ is constant.

3.2 Shock Region ($m_f(t) < m < m_s(t)$)

In the shocked region, the similarity variable is normalized to:

$$y_s = \frac{m - m_f(t)}{m_s(t) - m_f(t)} \in [0, 1] \tag{9}$$

Density Profile $\rho(m,t)$: The density increases from the piston boundary towards the shock front due to compression:

$$\rho(m,t) \approx 173.88\, y_s^{0.647468}\,\text{g/cm}^3 \tag{10}$$

At the shock front $y_s = 1$, the density reaches the strong shock Hugoniot limit of $\rho_s = 9\rho_0 = 173.88\,\text{g/cm}^3$.

Pressure Profile $p(m,t)$: The pressure profile scales as:

$$p(m,t) \approx 2.710000 \left(\frac{t}{\mathrm{ns}}\right)^{-0.447} \left[1.0 + 0.493384\, y_s^{1.133029}\right] \text{ MBar} \tag{11}$$

At the piston boundary $y_s = 0$, the pressure is precisely matched to the subsonic ablation pressure $p(m_f, t) \approx 2.71\,(t/\mathrm{ns})^{-0.447}$ MBar.

Velocity Profile $u(m,t)$: The velocity satisfies:

$$u(m,t) \approx \left(\frac{t}{\mathrm{ns}}\right)^{-0.2235} \left[3.65776 + 0.65734\, y_s^{0.644937}\right] \text{ km/s} \tag{12}$$

At the piston boundary $y_s = 0$, $u(m_f, t) \approx 3.65776\,(t/\mathrm{ns})^{-0.2235}$ km/s.

Temperature Profile $T(m,t)$: Using the ideal gas relation $T = p/((\gamma - 1)\rho)$ in HeV:

$$T(m,t) \approx 3.2091 \times 10^5 \left(\frac{t}{\mathrm{ns}}\right)^{-0.447} y_s^{-0.438483}\, \text{HeV} \tag{13}$$

Note the temperature singularity at the piston boundary $y_s \to 0$ due to the vanishing density.